



Fig. 3: Different feel due to surface for different object or image projected

particular challenges that must be addressed for widespread adoption. Technical hurdles, such as ensuring seamless integration with existing software platforms and optimising performance across different hardware configurations, remain significant barriers to overcome. Additionally, privacy and data security concerns must be carefully addressed to safeguard user information in immersive environments.

Looking ahead, the future of ITAD technology promises ongoing innovation. Advancements in display technologies, such as holographic displays and volumetric projections, may further enhance the realism and immersion of ITAD experiences. Furthermore, advancements in artificial intelligence and machine learning could enable more intelligent and context-aware

interactions within virtual environments, elevating the sophistication of ITAD interfaces.

ITAD technology represents a revolutionary approach to human-computer interaction, offering users a three-dimensional experience of touchscreen interaction that blurs the boundaries between the physical and digital worlds. With applications spanning education, healthcare, gaming, design, retail, and beyond, ITAD technology has the potential to transform how we learn, work, play, and communicate in the digital age. As research and development in this field continue to advance, we anticipate a future where immersive touch experiences become seamlessly integrated into our daily lives, unlocking new possibilities for creativity, collaboration, and innovation. **EFY**

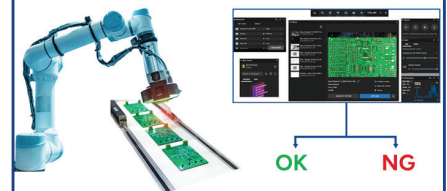
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HIGH-QUALITY OPTICAL INSPECTION SYSTEM IN MANUFACTURING FOR PRODUCT SORTING **OK / NG**

JB INSTRUMENTS Vision System's camera capturing a digital image using an image sensor. Once the image is captured, the vision system moves to processing and analysis. This stage is crucial for extracting useful information from the image.

Image Processing and Analysis

Feature Extraction: Key features, such as edges, corners, and textures, are identified and extracted from the image. These features are critical for understanding the object's shape, structure, and surface characteristics.



Decision Making and Output

After processing and analysing the image, the vision system makes decisions based on the results. This final stage is crucial for ensuring that the inspected objects meet the required standards and for maintaining the efficiency of the production line.

OK/NG Decision: The system determines if the inspected object meets required standards and mark it as PASS or FAIL. This decision is based on the criteria set for the specific application, ensuring that only compliant products proceed.

Pattern Recognition: Algorithms are used to recognize patterns and compare them with predefined templates to identify defects or ensure correct assembly. This process ensures precision and efficiency in various industrial applications, from quality control to automation.

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